

First name:

Last name:

Student ID:

Name and Student ID must be filled in for this page to be graded.

Problem 1. Find $\vec{\nabla} f$ and its value at P for

$$f = \frac{x}{x^2 + y^2}, \quad P(1, 1).$$

Solution.

□

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Problem 2. Find $\operatorname{div} \vec{v}$ and its value at P for

$$\vec{v} = \begin{bmatrix} 0 \\ \cos(xyz) \\ \sin(xyz) \end{bmatrix}, \quad P(2, \frac{\pi}{2}, 0).$$

Solution.

□